EXERCISE 6

Amplitude-Phase Transformation for Harmonic Oscillations

Original statement · ProblemsCh3.pdf, p. 2

Polar Representation • Phasor Geometry

PHASE 1: STATEMENT & SETUP

> Given an oscillating function of the form $x(t) = A\cos(\omega t) + B\sin(\omega t)$, show that it can be rewritten as a "single oscillation" $x(t) = M\cos(\omega t - \phi)$ with amplitude $M = (A^2 + B^2)^{1/2}$ and phase $\phi = \arctan(B/A)$.

PHASE 2: FUNDAMENTAL FORMULATION

Apply the angle subtraction identity $\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$ with $\alpha = \omega t$ and $\beta = \phi$, and equate coefficients with the Cartesian basis $\{\cos\omega t, \sin\omega t\}$.

> CLASS NOTES AND COMPLEMENT [NOTES, P. 5]
> Case 03 of the notes builds the real form from $C_1 e^{i\omega t} + C_2 e^{-i\omega t}$ with $C_1 = \frac{1}{2}(A - iB)$, $C_2 = \overline{C_1}$ (here $\rho = 0$). Writing $A = M\cos\phi$, $B = M\sin\phi$ gives $C_1 = \frac{1}{2}Me^{-i\phi}$, and the sum is $\mathrm{Re}\left(Me^{i(\omega t - \phi)}\right) = M\cos(\omega t - \phi)$: the amplitude-phase form of this problem, which is a complement (not in the class notes).

PHASE 3: STEP-BY-STEP ANALYTICAL DERIVATION

Step 1: Expand the single oscillation form:

$$x(t) = M\cos(\omega t - \phi) = M\left[\cos(\omega t)\cos\phi + \sin(\omega t)\sin\phi\right] = (M\cos\phi)\cos(\omega t) + (M\sin\phi)\sin(\omega t)$$

Step 2: Equate with Cartesian components:

$$A\cos(\omega t) + B\sin(\omega t) = (M\cos\phi)\cos(\omega t) + (M\sin\phi)\sin(\omega t)$$

$$\begin{cases} A = M\cos\phi & (1) \\ B = M\sin\phi & (2) \end{cases}$$

Step 3: Solve for Amplitude M:

Square and add (1) and (2):

$$A^2 + B^2 = M^2\cos^2\phi + M^2\sin^2\phi = M^2(\cos^2\phi + \sin^2\phi) = M^2$$

$$\mathbf{M = \sqrt{A^2 + B^2} = (A^2 + B^2)^{1/2}}$$

Step 4: Solve for Phase Lag ϕ:

Divide (2) by (1):

$$\frac{B}{A} = \frac{M\sin\phi}{M\cos\phi} = \tan\phi \implies \phi = \arctan\left(\frac{\mathbf{B}}{\mathbf{A}}\right)$$

PHASE 4: INTERPRETATION & ENGINEERING RELEVANCE

POLAR HARMONIC FORMULAS

$$\boxed{x(t) = M\cos(\omega t - \phi), \qquad M = \sqrt{A^2 + B^2}, \qquad \phi = \arctan(B/A)}$$

STRUCTURAL DYNAMICS APPLICATION

In aircraft wing flutter and structural modal analysis, the amplitude-phase form immediately reveals the maximum physical peak deflection $x_{\max} = M$ and time lag $\Delta t = \phi/\omega$ relative to external periodic gusts.